

Sinclair ZX80/ZX81 Emulator User Guide

ZX80 AND ZX81 ROM VARIANTS: STARTUP OPTIONS AND
COMMAND CONSOLE REFERENCE

IGNACIO CEA FORNIÉS

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Foreword

This guide describes the startup and command interfaces implemented by the two Sinclair ZX80/ZX81 executables in EMULATORS: ZX81Emulator (without a local command console) and ZX81EmulatorC (with a local command console). It covers the ZX80 and the three ZX81 ROM selectors and is based on both executable entry points, the ZX81::ZX81Emulator inheritance chain, LocalConsole, and the complete ZX81 -> SINCLAIR -> StandardCommandBuilder responsibility chain.

Command and parameter names are shown exactly as accepted by the current code. Startup options use /XVALUE with no space between the one-letter option and its value. Command names are matched as written by the builders; use uppercase names for reliable operation. Positional command parameters are separated by spaces; examples use paths without embedded spaces.

The non-console executable does not expose an interactive prompt. When started with /o it creates a remote communication system using the same ZX80/ZX81, Sinclair and standard command builders. Commands marked Local console only are intercepted by LocalConsole and therefore cannot be sent to the non-console executable.

1. Startup Parameters: Version Without a Local Console

Executable: ZX81Emulator.exe. It accepts the framework and ZX80/ZX81-specific options below, plus /o and the communication-port option. Option letters are case-sensitive because the command-line parser stores the character exactly as supplied.

/o

Meaning: Controls creation of the remote command communication system.

General form:

```
/o[YES|NO]
```

- YES or an empty value: open remote communications.
- NO (and any other non-YES non-empty value): do not open it.
- Omitted: remote communications disabled.

Examples:

```
ZX81Emulator.exe /o  
ZX81Emulator.exe /oYES /p60000
```

Consequence: When enabled, remote clients can execute ZX80/ZX81, Sinclair and standard builder commands. LocalConsole-only commands remain unavailable.

/p

Meaning: Selects the TCP listening port used by the non-console executable's remote command channel.

General form:

```
/pPORTNUMBER
```

- 0 through 65535: requested listening port; values above 1000 are recommended.
- Omitted: port 60000 is used when /o enables the channel.
- Collision: the inherited emulator also interprets /p as a comma-separated peripheral-ID list. A supplied port number is therefore also attempted as a peripheral ID during initialization.

Examples:

```
ZX81Emulator.exe /o  
ZX81Emulator.exe /o /p61000
```

Consequence: The remote server listens on the selected port. When /p is present, the same value is also passed to peripheral startup and normally produces a log warning because a port number is not a valid peripheral ID.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
ZX81Emulator.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
ZX81Emulator.exe /fprogram.bin  
ZX81Emulator.exe /fgames\demo.bin /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
ZX81Emulator.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
ZX81Emulator.exe /cstartup.asm /s
```


Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
ZX81Emulator.exe /l0  
ZX81Emulator.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: ZX80/ZX81 address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
ZX81Emulator.exe /a$080d  
ZX81Emulator.exe /fprogram.bin /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
ZX81Emulator.exe /d$080d  
ZX81Emulator.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
ZX81Emulator.exe /s  
ZX81Emulator.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the language identifier exposed to the emulated machine.

General form:

```
/iLANG
```

- ENG: English; this is the only language advertised and supported by the ZX80/ZX81 emulator.

Examples:

```
ZX81Emulator.exe /iENG
```

Consequence: The machine starts with the English language identifier. Other values are passed through but have no documented ZX80/ZX81 ROM variant in this implementation.

/r

Meaning: Requests the initial sound-output state.

General form:

```
/r[YES|NO]
```

- YES or an empty value: requests sound enabled.
- NO (and any other non-YES non-empty value): requests sound disabled.
- Omitted: requests sound enabled.

Examples:

```
ZX81Emulator.exe /r  
ZX81Emulator.exe /rNO
```

Consequence: The option is parsed, but the implemented ZX80/ZX81 computer has no sound device, so initialization makes no audio-state change.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
ZX81Emulator.exe /g  
ZX81Emulator.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CRTOFF can change it later.

/n

Meaning: Requests NTSC visualization parameters.

General form:

```
/n
```

- No value: the presence of the option requests NTSC.

Examples:

```
ZX81Emulator.exe /mZX813 /n
```

Consequence: The current factory always constructs PAL ZX80 and ZX81 machines. The option is detected and a limitation is written to the log, but PAL timing and screen geometry remain active.

/w

Meaning: Selects the startup RAM configuration.

General form:

```
/wCONF
```

- ZX80: only configuration 0 (unexpanded) is implemented; another value is ignored and logged.
- ZX81 ROM variants, 0 or omitted: unexpanded configuration (default).
- ZX81 ROM variants, 1: 16 KiB expansion.
- ZX81 ROM variants, any value other than 0 or 1: reset to unexpanded configuration and logged.

Examples:

```
ZX81Emulator.exe /mZX80 /w0  
ZX81Emulator.exe /mZX813 /w1
```

Consequence: The selected memory map is installed before the ROM and machine initialize. The broader 3K/8K/24K list in an old header comment is not implemented by the current factory.

/b

Meaning: Draws a diagnostic grid around the writable screen area and selects its color.

General form:

```
/b[COLOR]
```

- COLOR omitted or 0: black (default).
- 1 through 15: the requested grid color.
- Any value greater than 15: clamped to 15.

Examples:

```
ZX81Emulator.exe /b  
ZX81Emulator.exe /b14
```

Consequence: The screen is initialized with the diagnostic grid enabled in the selected color.

/m

Meaning: Selects the ZX80 or ZX81 ROM variant to emulate.

General form:

```
/mMACHINE
```

- ZX80: ZX80 model and ROM; this is the default when /m is omitted.
- ZX811: ZX81 with the old ROM image.
- ZX812: ZX81 with the rare/intermediate ROM image.
- ZX813: ZX81 with the newest ROM image.
- Any other value: logs the unsupported selector and falls back to ZX80.

Examples:

```
ZX81Emulator.exe /mZX80  
ZX81Emulator.exe /mZX811  
ZX81Emulator.exe /mZX813 /w1
```

Consequence: The selector chooses the ROM image and machine type created before initialization. It does not independently select a ULA silicon revision, video standard or RAM expansion.

2. Startup Parameters: Version With a Local Console

Executable: ZX81EmulatorC.exe. Its main program delegates the complete command line to ZX81::ZX81Emulator, so all inherited framework options and ZX80/ZX81-specific options are active even though the short main-level help banner only advertises /h. It has no /o listener option; commands are entered in its local console.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
ZX81EmulatorC.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
ZX81EmulatorC.exe /fprogram.bin  
ZX81EmulatorC.exe /fgames\demo.bin /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
ZX81EmulatorC.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
ZX81EmulatorC.exe /cstartup.asm /s
```

Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
ZX81EmulatorC.exe /l0  
ZX81EmulatorC.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: ZX80/ZX81 address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
ZX81EmulatorC.exe /a$080d  
ZX81EmulatorC.exe /fprogram.bin /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
ZX81EmulatorC.exe /d$080d  
ZX81EmulatorC.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
ZX81EmulatorC.exe /s  
ZX81EmulatorC.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the language identifier exposed to the emulated machine.

General form:

```
/iLANG
```

- ENG: English; this is the only language advertised and supported by the ZX80/ZX81 emulator.

Examples:

```
ZX81EmulatorC.exe /iENG
```

Consequence: The machine starts with the English language identifier. Other values are passed through but have no documented ZX80/ZX81 ROM variant in this implementation.

/r

Meaning: Requests the initial sound-output state.

General form:

```
/r[YES|NO]
```

- YES or an empty value: requests sound enabled.
- NO (and any other non-YES non-empty value): requests sound disabled.
- Omitted: requests sound enabled.

Examples:

```
ZX81EmulatorC.exe /r  
ZX81EmulatorC.exe /rNO
```

Consequence: The option is parsed, but the implemented ZX80/ZX81 computer has no sound device, so initialization makes no audio-state change.

/p

Meaning: Connects one or more peripherals during startup.

General form:

```
/pPER1[,PER2...]
```

- 10: typewriter keyboard input.
- 100: Sinclair cassette signal simulation.
- 101: ZX80/ZX81 cassette-data injection.
- 102: ZX Printer thermal-printer simulation.
- 201: cartridge implementation supported by the builder, although it is not listed by POSSIBLEPERS.

Examples:

```
ZX81EmulatorC.exe /p10  
ZX81EmulatorC.exe /p100,102
```

Consequence: Each requested peripheral is built and connected to the first compatible device during emulator initialization. Unsupported IDs are logged and initialization continues.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
ZX81EmulatorC.exe /g  
ZX81EmulatorC.exe /gNO
```


Consequence: The screen starts with or without CRT emulation. CRTON and CROFF can change it later.

/n

Meaning: Requests NTSC visualization parameters.

General form:

```
/n
```

- No value: the presence of the option requests NTSC.

Examples:

```
ZX81EmulatorC.exe /mZX813 /n
```

Consequence: The current factory always constructs PAL ZX80 and ZX81 machines. The option is detected and a limitation is written to the log, but PAL timing and screen geometry remain active.

/w

Meaning: Selects the startup RAM configuration.

General form:

```
/wCONF
```

- ZX80: only configuration 0 (unexpanded) is implemented; another value is ignored and logged.
- ZX81 ROM variants, 0 or omitted: unexpanded configuration (default).
- ZX81 ROM variants, 1: 16 KiB expansion.
- ZX81 ROM variants, any value other than 0 or 1: reset to unexpanded configuration and logged.

Examples:

```
ZX81EmulatorC.exe /mZX80 /w0  
ZX81EmulatorC.exe /mZX813 /w1
```

Consequence: The selected memory map is installed before the ROM and machine initialize. The broader 3K/8K/24K list in an old header comment is not implemented by the current factory.

/b

Meaning: Draws a diagnostic grid around the writable screen area and selects its color.

General form:

```
/b[COLOR]
```

- COLOR omitted or 0: black (default).
- 1 through 15: the requested grid color.
- Any value greater than 15: clamped to 15.

Examples:

```
ZX81EmulatorC.exe /b  
ZX81EmulatorC.exe /b14
```

Consequence: The screen is initialized with the diagnostic grid enabled in the selected color.

/m

Meaning: Selects the ZX80 or ZX81 ROM variant to emulate.

General form:

```
/mMACHINE
```

- ZX80: ZX80 model and ROM; this is the default when /m is omitted.
- ZX811: ZX81 with the old ROM image.
- ZX812: ZX81 with the rare/intermediate ROM image.
- ZX813: ZX81 with the newest ROM image.
- Any other value: logs the unsupported selector and falls back to ZX80.

Examples:

```
ZX81EmulatorC.exe /mZX80  
ZX81EmulatorC.exe /mZX811  
ZX81EmulatorC.exe /mZX813 /w1
```

Consequence: The selector chooses the ROM image and machine type created before initialization. It does not independently select a ULA silicon revision, video standard or RAM expansion.

3. Command Console Reference

The local console checks its eleven emulator-level commands first. All remaining text is delegated through ZX81::CommandBuilder, then SINCLAIR::CommandBuilder, and finally MCHEmul::StandardCommandBuilder. The remote /o channel in ZX81Emulator.exe uses only that three-builder chain. The availability line in each entry makes this boundary explicit.

3.1 Standard Framework Commands

HELP

Availability: Local console and remote /o channel.

Accepted aliases: CHELP.

Meaning: Displays the ZX80/ZX81 console help assembled from the ZX80/ZX81, Sinclair, local-console and standard help files.

General syntax:

```
HELP
```

Parameters:

- No parameters.

Examples:

```
HELP
```

Result and consequences: Displays the ZX80/ZX81 console help assembled from the ZX80/ZX81, Sinclair, local-console and standard help files.

AUTHOR

Availability: Local console and remote /o channel.

Accepted aliases: CAUTHOR.

Meaning: Returns framework author information.

General syntax:

```
AUTHOR
```

Parameters:

- No parameters.

Examples:

```
AUTHOR
```

Result and consequences: Returns framework author information.

STATUS

Availability: Local console and remote /o channel.

Accepted aliases: CSTATUS.

Meaning: Shows the Z80 status register.

General syntax:

```
STATUS
```

Parameters:

- No parameters.

Examples:

```
STATUS
```

Result and consequences: Shows the Z80 status register.

PC

Availability: Local console and remote /o channel.

Accepted aliases: CPC.

Meaning: Shows the current program counter.

General syntax:

```
PC
```

Parameters:

- No parameters.

Examples:

```
PC
```

Result and consequences: Shows the current program counter.

REGISTERS

Availability: Local console and remote /o channel.

Accepted aliases: CREGISTERS.

Meaning: Shows all CPU registers.

General syntax:

```
REGISTERS
```

Parameters:

- No parameters.

Examples:

```
REGISTERS
```

Result and consequences: Shows all CPU registers.

CHANGereg

Availability: Local console and remote /o channel.

Accepted aliases: CCHANGereg.

Meaning: Changes one CPU internal register.

General syntax:

```
CHANGereg REGNAME VALUE
```

Parameters:

- REGNAME: registered Z80 register name.
- VALUE: new decimal, z-prefixed binary, o-prefixed octal or \$-prefixed hexadecimal value.

Examples:

```
CHANGereg A $41
```

Result and consequences: The register changes only if it exists and accepts the supplied byte width; otherwise an error is returned.

STACK

Availability: Local console and remote /o channel.

Accepted aliases: CSTACK.

Meaning: Shows stack state.

General syntax:

```
STACK [ALL]
```

Parameters:

- ALL: optional; includes stack contents in addition to basic state.

Examples:

```
STACK  
STACK ALL
```

Result and consequences: Shows stack state.

CPUSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATUS.

Meaning: Shows registers, status, program counter and stack.

General syntax:

```
CPUSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSTATUS
```

Result and consequences: Shows registers, status, program counter and stack.

CPUSSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSSTATUS.

Meaning: Shows the compact CPU status without stack detail.

General syntax:

```
CPUSSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSSTATUS
```

Result and consequences: Shows the compact CPU status without stack detail.

CPUSTATE

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATE.

Meaning: Shows the CPU execution state and state-specific data.

General syntax:

```
CPUSTATE
```

Parameters:

- No parameters.

Examples:

```
CPUSTATE
```

Result and consequences: Shows the CPU execution state and state-specific data.

CPUINFO

Availability: Local console and remote /o channel.

Accepted aliases: CCPUINFO.

Meaning: Returns the complete CPU information structure.

General syntax:

```
CPUINFO
```

Parameters:

- No parameters.

Examples:

```
CPUINFO
```

Result and consequences: Returns the complete CPU information structure.

MEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORY.

Meaning: Reads the active CPU-visible memory range.

General syntax:

```
MEMORY INITIAL [FINAL]
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address; with no final address only the initial location is reported.

Examples:

```
MEMORY $0400  
MEMORY $0400 $0427
```

Result and consequences: Reads the active CPU-visible memory range.

MEMORYDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYDUMP.

Meaning: Dumps every memory view between two addresses, including inactive RAM/ROM views.

General syntax:

```
MEMORYDUMP INITIAL FINAL
```

Parameters:

- INITIAL: first address.
- FINAL: last address.

Examples:

```
MEMORYDUMP $a000 $a03f
```

Result and consequences: Dumps every memory view between two addresses, including inactive RAM/ROM views.

MEMORYSTR

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYSTR.

Meaning: Shows the configured memory structure.

General syntax:

```
MEMORYSTR [ALL]
```

Parameters:

- ALL: optional; includes inactive regions. Without it only active regions are shown.

Examples:

```
MEMORYSTR  
MEMORYSTR ALL
```

Result and consequences: Shows the configured memory structure.

SETMEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CSETMEMORY.

Meaning: Writes a value to one address or repeatedly across a range.

General syntax:

```
SETMEMORY INITIAL [FINAL] VALUE
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address.
- VALUE: byte sequence in framework numeric syntax; multi-byte values advance by their own length.

Examples:

```
SETMEMORY $0400 $41  
SETMEMORY $0400 $0427 $20
```

Result and consequences: The active memory view is modified immediately; unsafe writes can corrupt state or crash the emulated program.

MEMORYANALYSIS

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYANALYSIS.

Meaning: Analyzes the current memory configuration.

General syntax:

```
MEMORYANALYSIS
```

Parameters:

- No parameters.

Examples:

```
MEMORYANALYSIS
```

Result and consequences: Analyzes the current memory configuration.

STOP

Availability: Local console and remote /o channel.

Accepted aliases: CSTOP.

Meaning: Stops CPU execution.

General syntax:

```
STOP
```

Parameters:

- No parameters.

Examples:

```
STOP
```

Result and consequences: Stops CPU execution.

RUN

Availability: Local console and remote /o channel.

Accepted aliases: R, CRUN.

Meaning: Starts or resumes CPU execution; alias R.

General syntax:

```
RUN [ADDRESS]
```

Parameters:

- ADDRESS: optional new program counter before running.

Examples:

```
RUN  
R $080d
```

Result and consequences: The CPU resumes immediately, optionally after changing PC.

SETPC

Availability: Local console and remote /o channel.

Accepted aliases: CSETPC.

Meaning: Sets the program counter without starting execution.

General syntax:

```
SETPC ADDRESS
```

Parameters:

- ADDRESS: new CPU address.

Examples:

```
SETPC $080d
```

Result and consequences: Subsequent execution continues from the new address; an unsafe address can break the program.

NEXT

Availability: Local console and remote /o channel.

Accepted aliases: N, CNEXT.

Meaning: Executes the next instruction; alias N.

General syntax:

```
NEXT
```

Parameters:

- No parameters.

Examples:

```
NEXT  
N
```

Result and consequences: Executes the next instruction; alias N.

NEXTSTACK

Availability: Local console and remote /o channel.

Accepted aliases: NS, CNEXTSTACK.

Meaning: Runs until the stack pointer returns to its initial value; alias NS.

General syntax:

```
NEXTSTACK
```

Parameters:

- No parameters.

Examples:

```
NS
```

Result and consequences: Useful for stepping over a routine, although behavior depends on the routine's stack discipline.

SHOWNEXT

Availability: Local console and remote /o channel.

Accepted aliases: SN, CSHOWNEXT.

Meaning: Disassembles upcoming instruction(s); alias SN.

General syntax:

```
SHOWNEXT [NUMBER]
```

Parameters:

- NUMBER: optional count; one instruction by default.

Examples:

```
SHOWNEXT  
SN 8
```

Result and consequences: Disassembles upcoming instruction(s); alias SN.

INST

Availability: Local console and remote /o channel.

Accepted aliases: CINST.

Meaning: Shows the last instruction executed.

General syntax:

```
INST
```

Parameters:

- No parameters.

Examples:

```
INST
```

Result and consequences: Shows the last instruction executed.

BREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CBREAKS.

Meaning: Lists all stop-point addresses.

General syntax:

```
BREAKS
```

Parameters:

- No parameters.

Examples:

```
BREAKS
```

Result and consequences: Lists all stop-point addresses.

SETBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CSETBREAK.

Meaning: Adds one or more breakpoints.

General syntax:

```
SETBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more addresses in framework numeric syntax.

Examples:

```
SETBREAK $080d $ffd2
```

Result and consequences: Adds one or more breakpoints.

REMOVEBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAK.

Meaning: Removes selected breakpoints.

General syntax:

```
REMOVEBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more existing breakpoint addresses.

Examples:

```
REMOVEBREAK $ffd2
```

Result and consequences: Removes selected breakpoints.

REMOVEBREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAKS.

Meaning: Removes every breakpoint.

General syntax:

```
REMOVEBREAKS
```

Parameters:

- No parameters.

Examples:

```
REMOVEBREAKS
```

Result and consequences: Removes every breakpoint.

SPEED

Availability: Local console and remote /o channel.

Accepted aliases: CSPEED.

Meaning: Shows measured CPU and screen-I/O execution speed.

General syntax:

```
SPEED
```

Parameters:

- No parameters.

Examples:

```
SPEED
```

Result and consequences: Shows measured CPU and screen-I/O execution speed.

LOADBIN

Availability: Local console and remote /o channel.

Accepted aliases: CLOADBIN.

Meaning: Loads a raw binary file at an explicit address.

General syntax:

```
LOADBIN FILE ADDRESS
```

Parameters:

- FILE: raw binary path.
- ADDRESS: first destination address.

Examples:

```
LOADBIN charset.bin $2000
```

Result and consequences: Bytes are copied into memory without an embedded load-address header.

SAVEBIN

Availability: Local console and remote /o channel.

Accepted aliases: CSAVEBIN.

Meaning: Saves a memory range as raw binary.

General syntax:

```
SAVEBIN FILE SIZE ADDRESS
```

Parameters:

- FILE: output path.
- SIZE: byte count.
- ADDRESS: first source address.

Examples:

```
SAVEBIN screen.bin 1000 $0400
```

Result and consequences: Saves a memory range as raw binary.

DEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CACTIVATEDEEPDEBUG.

Meaning: Enables deep execution debugging.

General syntax:

```
DEBUGON FILE CPU [APPEND] [FILTERS...]
```

Parameters:

- FILE: output log.
- CPU: YES includes CPU execution; NO omits it.
- APPEND: optional YES appends, NO overwrites (default).
- D:START-END or ALLDIR: address filter.
- C:IDs or ALLCHIP: chip filter.
- I:IDs or ALLIO: device filter.
- M:IDs or ALLMEM: memory filter.

Examples:

```
DEBUGON zx81.log YES NO D:$0800-$0fff ALLCHIP ALLIO ALLMEM
```

Result and consequences: Tracing begins with the selected sources and can substantially slow emulation. It cannot coexist with interrupt-only debugging.

DEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CDESACTIVATEDEEPDEBUG.

Meaning: Stops deep execution debugging.

General syntax:

```
DEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
DEBUGOFF
```

Result and consequences: Stops deep execution debugging.

IDEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGON.

Meaning: Enables interrupt-only debugging.

General syntax:

```
IDEBUGON FILE [APPEND]
```

Parameters:

- FILE: output log.
- APPEND: optional YES appends; NO overwrites (default).

Examples:

```
IDEBUGON interrupts.log NO
```

Result and consequences: Interrupt tracing begins and cannot coexist with deep debugging.

IDEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGOFF.

Meaning: Stops interrupt-only debugging.

General syntax:

```
IDEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
IDEBUGOFF
```

Result and consequences: Stops interrupt-only debugging.

RESTART

Availability: Local console and remote /o channel.

Accepted aliases: CRESTART.

Meaning: Requests a restart and exits the current computer loop so it can be rebuilt.

General syntax:

```
RESTART STOPPED [LEVEL]
```

Parameters:

- STOPPED: YES restarts stopped; NO restarts running.
- LEVEL: optional 0 CPU-only/mild restart (default), 1 computer restart preserving memory, greater than 1 full restart.

Examples:

```
RESTART YES 0
RESTART NO 2
```

Result and consequences: Requests a restart and exits the current computer loop so it can be rebuilt.

DEVICES

Availability: Local console and remote /o channel.

Accepted aliases: CDEVICES.

Meaning: Shows all or selected I/O devices.

General syntax:

```
DEVICES [ID1 ID2 ...]
```

Parameters:

- IDn: optional device identifiers; with none, all devices are returned.

Examples:

```
DEVICES
DEVICES 1 2
```

Result and consequences: Shows all or selected I/O devices.

PERIPHERALS

Availability: Local console and remote /o channel.

Accepted aliases: CPERIPHERALS.

Meaning: Shows all or selected connected peripherals.

General syntax:

```
PERIPHERALS [ID1 ID2 ...]
```

Parameters:

- IDn: optional peripheral identifiers; with none, all are returned.

Examples:

```
PERIPHERALS
PERIPHERALS 100
```

Result and consequences: Shows all or selected connected peripherals.

PERCOMMAND

Availability: Local console and remote /o channel.

Accepted aliases: CPERCMD.

Meaning: Sends a device-specific command to a connected peripheral.

General syntax:

```
PERCOMMAND PERID CMDID [ATTRS...]
```

Parameters:

- PERID: connected peripheral ID.
- CMDID: command ID implemented by that peripheral.
- ATTRS: optional positional attributes forwarded unchanged.

Examples:

```
PERCOMMAND 100 0
```

Result and consequences: The peripheral executes the request if both IDs and its parameter contract are valid; otherwise the answer reports an error.

ASSIGNJOY

Availability: Local console and remote /o channel.

Accepted aliases: CASSIGNJ.

Meaning: Remaps a joystick identifier.

General syntax:

```
ASSIGNJOY ID NEWID
```

Parameters:

- ID: current joystick ID.
- NEWID: required target ID; using the same ID removes an existing assignment.

Examples:

```
ASSIGNJOY 0 1
```

Result and consequences: Remaps a joystick identifier.

CLOCKFACTOR

Availability: Local console and remote /o channel.

Accepted aliases: CCLOCKFACTOR.

Meaning: Changes the CPU clock multiplier.

General syntax:

```
CLOCKFACTOR VALUE
```

Parameters:

- VALUE: numeric factor; 1 means normal speed.

Examples:

```
CLOCKFACTOR 2
```

Result and consequences: The emulation pacing target changes; values above one request faster execution.

SOUNDON

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDON.

Meaning: Requests that sound output be enabled.

General syntax:

```
SOUNDON
```

Parameters:

- No parameters.

Examples:

```
SOUNDON
```

Result and consequences: The command is accepted, but the implemented ZX80/ZX81 computer has no sound device, so it produces no state change.

SOUNDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDOFF.

Meaning: Requests that sound output be disabled.

General syntax:

```
SOUNDOFF
```

Parameters:

- No parameters.

Examples:

```
SOUNDOFF
```

Result and consequences: The command is accepted, but the implemented ZX80/ZX81 computer has no sound device, so it produces no state change.

INTERRUPTS

Availability: Local console and remote /o channel.

Accepted aliases: CINTERRUPTS.

Meaning: Shows all interrupts known by the computer.

General syntax:

```
INTERRUPTS
```

Parameters:

- No parameters.

Examples:

```
INTERRUPTS
```

Result and consequences: Shows all interrupts known by the computer.

INTSET

Availability: Local console and remote /o channel.

Accepted aliases: CINTSET.

Meaning: Enables or disables one interrupt or every interrupt.

General syntax:

```
INTSET ID|ALL ON|OFF
```

Parameters:

- ID or ALL: target interrupt.
- ON or OFF: requested state.

Examples:

```
INTSET ALL ON
```

Result and consequences: An unknown interrupt ID produces an error.

CHIPS

Availability: Local console and remote /o channel.

Accepted aliases: CCHIPS.

Meaning: Lists the emulated chips and their identifying attributes.

General syntax:

```
CHIPS
```

Parameters:

- No parameters.

Examples:

```
CHIPS
```

Result and consequences: Lists the emulated chips and their identifying attributes.

CRTON

Availability: Local console and remote /o channel.

Accepted aliases: CCRTON.

Meaning: Enables the CRT screen effect.

General syntax:

```
CRTON
```

Parameters:

- No parameters.

Examples:

```
CRTON
```

Result and consequences: Enables the CRT screen effect.

CRTOFF

Availability: Local console and remote /o channel.

Accepted aliases: CCRTOFF.

Meaning: Disables the CRT screen effect.

General syntax:

```
CRTOFF
```

Parameters:

- No parameters.

Examples:

```
CRTOFF
```

Result and consequences: Disables the CRT screen effect.

DATASETTE

Availability: Local console and remote /o channel.

Accepted aliases: CDATASETTE.

Meaning: Shows datasette state when a supported datasette is connected.

General syntax:

```
DATASETTE
```

Parameters:

- No parameters.

Examples:

```
DATASETTE
```

Result and consequences: Shows datasette state when a supported datasette is connected.

GRIDON

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDON.

Meaning: Enables the diagnostic grid overlay on the screen.

General syntax:

```
GRIDON COLOR
```

Parameters:

- COLOR: mandatory numeric grid color.

Examples:

```
GRIDON 14
```

Result and consequences: Enables the standard screen grid with the requested color.

GRIDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDOFF.

Meaning: Disables the diagnostic grid overlay.

General syntax:

```
GRIDOFF
```

Parameters:

- No parameters.

Examples:

```
GRIDOFF
```

Result and consequences: Disables the standard screen grid.

PICTURE

Availability: Local console and remote /o channel.

Accepted aliases: CPICTURE.

Meaning: Saves a screenshot of the emulator screen.

General syntax:

```
PICTURE FILE
```

Parameters:

- FILE: output PNG, JPG or BMP filename.

Examples:

```
PICTURE zx81-screen.png
```

Result and consequences: A supported extension writes the current screen; unsupported formats fail.

SETHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CSETHOOK.

Meaning: Creates an execution hook.

General syntax:

```
SETHOOK ID TYPE [PARAMS...]
```

Parameters:

- ID: unique hook identifier.
- TYPE: hook kind; currently 0 stops on memory read/write.
- For TYPE 0, PARAMS are mandatory INITIAL and optional FINAL addresses.

Examples:

```
SETHOOK 1 0 $d000 $d3ff
```

Result and consequences: Creates an execution hook.

REMOVEHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEHOOK.

Meaning: Removes a hook by identifier.

General syntax:

```
REMOVEHOOK ID
```

Parameters:

- ID: hook identifier.

Examples:

```
REMOVEHOOK 1
```

Result and consequences: If the hook does not exist, no state changes.

HOOKS

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKS.

Meaning: Lists active hooks.

General syntax:

```
HOOKS
```

Parameters:

- No parameters.

Examples:

```
HOOKS
```

Result and consequences: Lists active hooks.

HOOKSHELP

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKSHELP.

Meaning: Shows help for supported hook types.

General syntax:

```
HOOKSHELP
```

Parameters:

- No parameters.

Examples:

```
HOOKSHELP
```

Result and consequences: Shows help for supported hook types.

CMOVEPARAMS

Availability: Local console and remote /o channel.

Accepted aliases: No public alias.

Meaning: Builds the framework's internal MoveParametersToAnswerCommand.

General syntax:

```
CMOVEPARAMS
```

Parameters:

- No parameters are executable under the current command contract.

Examples:

```
CMOVEPARAMS
```

Result and consequences: The command returns an empty answer when invoked directly; it is an internal response helper and is not advertised in the help file.

3.2 Inherited Sinclair Commands

SYSVARS

Availability: Local console and remote /o channel.

Accepted aliases: CSYSVARS.

Meaning: Returns every system variable defined for the selected ZX80 or ZX81 ROM profile.

General syntax:

```
SYSVARS
```

Parameters:

- No parameters.

Examples:

```
SYSVARS
```

Result and consequences: Returns the current values through the system-variable definitions loaded for the selected ROM variant.

SYSVAR

Availability: Local console and remote /o channel.

Accepted aliases: CSYSVAR.

Meaning: Returns one named system variable.

General syntax:

```
SYSVAR VNAME
```

Parameters:

- VNAME: mandatory variable name from the selected machine's system-variable table.

Examples:

```
SYSVAR D_FILE  
SYSVAR E_LINE
```

Result and consequences: Returns the named variable and its current memory value; an unknown name returns an error entry.

3.3 ZX80/ZX81-Specific Commands

ULA

Availability: Local console and remote /o channel.

Accepted aliases: CULA.

Meaning: Returns the current ULA and video-generation state.

General syntax:

```
ULA
```

Parameters:

- No parameters.

Examples:

```
ULA
```

Result and consequences: Returns the ULA state for the selected ZX80 or ZX81 ROM variant, including its current raster and display-generation data.

ULAEVENTS

Availability: Local console and remote /o channel.

Accepted aliases: CULAEVENTS.

Meaning: Enables or disables visualization of significant CPU/ULA events.

General syntax:

```
ULAEVENTS ON|OFF
```

Parameters:

- ON: enables event visualization.
- OFF: disables it. The implementation accepts any single token; only the exact value ON enables the feature.

Examples:

```
ULAEVENTS ON
ULAEVENTS OFF
```

Result and consequences: Changes the ULA event-overlay flag used by the active ZX80/ZX81 screen implementation.

DFDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CDFDUMP.

Meaning: Returns a snapshot of the current display file.

General syntax:

```
DFDUMP
```

Parameters:

- No parameters.

Examples:

```
DFDUMP
```

Result and consequences: Returns the bytes selected by the active ZX80/ZX81 display-file mapping.

CHARSDRAW

Availability: Local console and remote /o channel.

Accepted aliases: CCHARSDRAW.

Meaning: Returns textual drawings of all or selected character glyphs.

General syntax:

```
CHARSDRAW [0..63 ...]
```

Parameters:

- Character codes: optional values from 0 through 63; out-of-range values are ignored and duplicates are coalesced. With none, all 64 glyphs are rendered.

Examples:

```
CHARSDRAW  
CHARSDRAW 0 1 32 63
```

Result and consequences: Returns textual glyph drawings from the character data visible to the selected ZX80/ZX81 model.

3.4 Local-Console-Only Commands

LOADPRG

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads, assembles and installs an assembler source through the emulator object.

General syntax:

```
LOADPRG FILE
```

Parameters:

- FILE: assembler source path.

Examples:

```
LOADPRG demo.asm
```

Result and consequences: Compilation or loading errors are reported by the local console.

LOADBINARY

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a binary containing its destination address in the file header.

General syntax:


```
LOADBINARY FILE
```

Parameters:

- FILE: header-addressed binary path.

Examples:

```
LOADBINARY demo.bin
```

Result and consequences: Loads a binary containing its destination address in the file header.

LOADBLOCKS

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a block-structured memory file.

General syntax:

```
LOADBLOCKS FILE
```

Parameters:

- FILE: block file path.

Examples:

```
LOADBLOCKS state.blocks
```

Result and consequences: Loads a block-structured memory file.

DECOMPILE

Availability: Local console only.

Accepted aliases: None.

Meaning: Disassembles a memory region.

General syntax:

```
DECOMPILE ADDRESS BYTES
```

Parameters:

- ADDRESS: first address.
- BYTES: approximate byte count; output rounds to complete instructions.

Examples:

```
DECOMPILE $080d 64
```

Result and consequences: Disassembles a memory region.

POSSIBLEPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Lists the peripheral IDs and construction attributes advertised by the active ZX80/ZX81 I/O peripheral builder.

General syntax:

```
POSSIBLEPERS
```

Parameters:

- No parameters.

Examples:

```
POSSIBLEPERS
```

Result and consequences: Lists IDs 10 (typewriter), 100 (cassette), 101 (cassette injection) and 102 (thermal printer). Cartridge ID 201 is accepted by the builder but is not included in this advertised list.

CONNECTPER

Availability: Local console only.

Accepted aliases: None.

Meaning: Builds and connects a peripheral to the first compatible ZX80/ZX81 device.

General syntax:

```
CONNECTPER ID [ATTRS...]
```

Parameters:

- ID: peripheral-emulation ID (10, 100, 101, 102 or the additionally supported cartridge ID 201).
- ATTRS: optional construction attributes. For printer 102, use f:FILENAME and p:THERMAL or p:THERMAL-PS.

Examples:

```
CONNECTPER 100  
CONNECTPER 102 f:printer.ps p:THERMAL-PS
```

Result and consequences: Connection succeeds only when the ID, attributes and a compatible target device are available; otherwise the answer reports an error.

DISCONNECTPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Disconnects one or more peripherals.

General syntax:

```
DISCONNECTPERS ID1 [ID2 ...]
```

Parameters:

- IDn: connected peripheral IDs.

Examples:

```
DISCONNECTPERS 100 101
```

Result and consequences: Disconnects one or more peripherals.

LOADPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads media or program data into a connected peripheral.

General syntax:

```
LOADPERDATA ID FILE [DEBUGFILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: supported data file; for example sampled TZX data for ID 100 or an O/P program for injection ID 101.
- DEBUGFILE: optional file that starts full-range format-related debugging after a successful load.

Examples:

```
LOADPERDATA 100 tape.tzx  
LOADPERDATA 101 game.p load-trace.log
```

Result and consequences: The operation fails if the peripheral does not accept the supplied file type; otherwise the data is attached and optional debugging begins.

EMPTYPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Creates and attaches empty media data when the peripheral implements that operation.

General syntax:

```
EMPTYPERDATA ID FILE
```

Parameters:

- ID: connected peripheral ID.
- FILE: destination whose extension selects a writable format.

Examples:

```
EMPTYPERDATA 100 blank.tzx
```

Result and consequences: The Sinclair cassette implementation can create an empty TZX container; unsupported peripherals or formats return an error.

SAVEPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Saves a peripheral's current media data.

General syntax:

```
SAVEPERDATA ID [FILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: optional output path; it may be omitted when LOADPERDATA or EMPTYPERDATA established the current filename.

Examples:

```
SAVEPERDATA 100
SAVEPERDATA 100 copy.tzx
```

Result and consequences: Saves the current cassette data when the peripheral can retrieve it and the selected file format can be written.

CLEARPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Clears all data currently loaded into peripherals.

General syntax:

```
CLEARPERDATA
```

Parameters:

- No parameters.

Examples:

```
CLEARPERDATA
```

Result and consequences: Commonly used to force a subsequent program/media reload.

Appendices

Appendix A. Command Availability Summary

The table distinguishes the interactive console from the command channel of the non-console executable. Remote availability assumes ZX81Emulator.exe was started with /o and that the client is connected.

Command family	ZX81EmulatorC local console	ZX81Emulator remote /o
Standard builder	Yes	Yes
Inherited Sinclair builder	Yes	Yes
ZX80/ZX81 builder	Yes	Yes
LocalConsole emulator/media operations	Yes	No

Appendix B. Source-of-Truth Maintenance Rule

This file is a maintained repository artifact. Every change that adds, removes, renames or changes a ZX80/ZX81 startup option, a command-builder registration, a LocalConsole-only command, command syntax, parameter meaning, command consequences or user-visible formatted output must include a review and, when applicable, an update of docs/ZX81Data/ZX81Emulator_UserGuide.docx. The corresponding canonical .fmt source must be updated whenever the command's InfoStructure contract changes.

Audit the complete chain, not only the edited class: the ZX81Emulator entry point and ZX81::ZX81Emulator hierarchy for startup options; LocalConsole for local-only commands; and ZX81::CommandBuilder -> SINCLAIR::CommandBuilder -> MCHEmul::StandardCommandBuilder for builder commands.

Illustrations

NO ILLUSTRATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS

Equations

NO EQUATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS